

RICARDO CANNIZZARO

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OBJECTIVE

Causal ML researcher and robotics engineer | PhD candidate at the Oxford Robotics Institute (thesis submitted; awaiting defence) | Two-time Microsoft Research PhD Research Intern (Redmond) | Former Australian Defence Senior Research Scientist (autonomous aerial systems) | 10+ years experience designing and implementing AI/ML-enabled software systems for learning, inference, and decision-making under uncertainty in real-world, safety-critical and human-facing domains | Passionate about STEM education, outreach, diversity and inclusion.

Seeking senior research roles advancing human-centred and interactive AI systems, learning and reasoning with foundation and generative models, and robust, trustworthy decision-making under uncertainty in large-scale, real-world deployments.

Research Interests: Uncertainty-aware and probabilistic modelling, causal representation learning, and probabilistic programming; human-in-the-loop learning, interpretability and transparency, and counterfactual attribution and explanations; knowledge representation and reasoning, and autonomous systems assurance; multi-modal generative AI models (latent diffusion, transformers), with applications to human-facing agent decision-making and interaction in complex, uncertain, mixed virtual/real-world environments.

EDUCATION

DPhil (PhD) Engineering Science, Oxford Robotics Institute, University of Oxford. *Thesis submitted; awaiting defence (2026)*
Supervisors: Prof Lars Kunze & Prof Nick Hawes (Cognitive Robotics Group;
Goal-Oriented Long-Lived Systems Group). Funded by Defence Science & Technology Group Australia.

Thesis: *Causal Artificial Intelligence for Robust Robot Reasoning under Uncertainty*

PhD research on uncertainty-aware and probabilistic causal modelling for learning, inference, decision-making, and explanation in complex, partially observable environments. My work focuses on learning human-aligned causal representations and producing faithful, counterfactual-based explanations to support understanding, trust, and decision-making by non-technical users. Methodologically, this includes causal representation learning, Bayesian inference, and probabilistic programming, applied across both software-based virtual environments and real-world, hardware-integrated systems. This research included, and subsequently extended, work completed through two Microsoft Research PhD internships, each forming a core thesis chapter: one on counterfactual consistency for image fine-tuning, and one on formalising parallel-world consistency to enable generative models to learn and reason about video game mechanics and environment dynamics beyond pixel-level generation.

Courses: Oxford Scientific Entrepreneurship Course; Oxford Language Centre Italian Fast-Track Course (Parts 1–2)

Bachelor of Engineering (Robotics & Mechatronics) (Hons), Swinburne University, Australia *2013–2016*
First Class Honours (GPA 4.0/4.0). Highest achieving graduate. Industry-based learning year with DSTG.

Bachelor of Science (Mechatronic Systems), University of Melbourne, Australia *2010–2012*
Included a six-month international exchange at KTH Royal Institute of Technology, Stockholm, Sweden.

WORK EXPERIENCE

PhD Research Intern (Advanced Research Internship) May 2025 – Aug 2025
Microsoft Research (MSR) *Redmond, WA, USA*

- Conducted advanced research on formalising causal and consistency constraints for foundation and generative models in interactive virtual environments, enabling models to learn environment dynamics and video game mechanics beyond pixel-level generation

- Developed parallel-world consistency frameworks allowing generative models to reason over multiple hypothetical trajectories and counterfactual outcomes in structured environments
- Designed and implemented an instrumented interactive environment (using PyGame) that functions both as a configurable battle simulation and as a playable battle game, enabling controlled experimentation, data generation, and human interaction within the same system
- Used this environment to generate structured training data with explicit parallel-world and counterfactual consistency guarantees, supporting mechanics-aware learning and principled evaluation of generative models
- Implemented and trained deep generative models using PyTorch and Pyro probabilistic programming language, integrating causal constraints into modern Transformer-based architectures via the HuggingFace ecosystem
- Integrated causal modelling principles with end-to-end generative learning pipelines to support both learning and generation in complex simulated environments
- Produced a first-author research paper (*Multiverse Mechanics*), submitted to ICLR 2026 (currently under review)
- Delivered an end-of-internship research presentation to the AI Interaction and Learning Group at MSR, receiving strong positive feedback on both the scientific contribution and potential product impact of the work
- Created a research poster and presented the work at the 2025 MSR Intern Redmond Showcase Event to Microsoft and MSR staff, and fellow interns
- Presented live demonstrations of the system to internal Microsoft audiences including Xbox Gaming, Activision Blizzard, and Mojang, receiving strong feedback and an invitation for a follow-up extended technical presentation
- Worked under the mentorship of Dr Robert Ness at the AI Interaction and Learning Group at MSR

PhD Research Intern
Microsoft Research (MSR)

May 2024 – Aug 2024
Redmond, WA, USA

- Conducted foundational research on addressing causal reasoning and counterfactual consistency limitations in multi-modal generative AI models, with a focus on human-aligned image fine-tuning
- Developed and evaluated novel counterfactual consistency objectives for diffusion-based models, drawing on causal ML principles and representation learning
- Implemented new model architectures, loss functions, and training pipelines using PyTorch, Pyro probabilistic programming language, HuggingFace Transformers, and Weights & Biases
- Performed qualitative analysis of generated outputs to compare proposed methods against baseline generative models and assess counterfactual consistency behaviour
- Communicated research progress through structured mid-point and final research presentations
- Advocated for diversity, equity, and inclusion (DEI) by proactively participating in DEI initiatives, including speaking on a MSR internship DEI panel
- Worked under the mentorship of Dr Robert Ness and Dr Emre Kiciman in the AI for Industries and AI Resilience research groups at MSR

Senior Robotics Specialist / Senior Defence Research Scientist
Australian Defence Science and Technology Group (DSTG)

July 2015 – Nov 2025
Melbourne, Australia

- Conducted applied research on trusted autonomy for aerial and multi-robot systems in safety-critical environments, delivering scientific and capability-oriented impact for Defence and industry partners as research clients
- Served as a key research contributor to the Defence-led Cooperative Research Centre (CRC) for Trusted Autonomous Systems, collaborating with industry partners including DefendTex and Mission Systems to co-develop research-driven autonomy solutions. This work supported industry-led development of deployable autonomous systems that were adopted by Defence customers and strengthened the local Defence innovation ecosystem.
- Led and contributed to the design, integration, and deployment of scalable multi-robot exploration and return systems for operation in a priori unknown environments across simulation and real-world platforms

- Contributed to international collaborative Defence research programs (including the United States and United Kingdom) focused on human-in-the-loop command and control and common operating picture (C2/COP) software for real-time tasking of autonomous drone systems
- Developed and evaluated autonomy-enabled workflows that generate ISR products such as maps of unknown environments on demand, supporting ISR analysts and human decision-makers in operational planning and situational awareness
- Undertook deputy manager responsibilities for the Fishermans Bend Flight Laboratory, including laboratory operations, safety compliance, staff training, and maintenance of critical research infrastructure
- Published peer-reviewed research at leading international conferences (e.g. ICRA, ACRA) on multi-robot coordination, navigation, SLAM, and decision-making under uncertainty
- Supported and delivered large-scale demonstrations and field trials of autonomous systems in Australia, the United States, Canada, and Singapore in collaboration with government, industry, and academic partners

PROJECTS

- *Multiverse Mechanics: A Benchmark for Learning Game Mechanics via Counterfactual Worlds* [\[Paper\]](#)
- Counterfactual Contrastive Learning for Improving Causal Consistency in Multi-Modal GenAI
- Causal Reasoning & Explanations for Robot Manipulation: [\[Project\]](#)
- Causal Discovery, Inference & Counterfactual Explanations for Drones in Mines: [\[Project\]](#)
- Causally-Informed POMDP Planning for Robots Under Confounded Decision-Making: [\[Project\]](#)
- Scaleable Multi-Robot Exploration & Return in Apriori Unknown Environments: [\[Project\]](#)
- Decentralised MCTS for Robot Intelligence, Surveillance & Reconnaissance Missions: [\[Project\]](#)
- Team ORIon Team Lead (2021/22): [\[Project\]](#)
- Team ORIon Task-Level Planning Sub-Team Lead (2021/22): [\[Project\]](#)
- Autonomous Drone Navigation & Mapping in Mixed GNSS-Available Environments: [\[Project\]](#)
- Random-Finite-Set-based SLAM algorithm for drones with scanning & solid-state LIDARS: [\[Project\]](#)

AWARDS & FUNDING

- ECMR 2025 Overall Best Conference Paper Award [\[Announcement\]](#)
- Invited to contribute a full journal paper to the *Robotics and Autonomous Systems* ECMR 2025 special issue
- USD \$250 IEEE Robot Learning TC travel grant for the IROS 2023 [Causality for Robotics workshop](#)
- USD \$305 IEEE Robotics and Automation Society (RAS) member support program grant for IROS 2023
- £340 St Edmund Hall Postgraduate College Grant for IROS 2023 conference attendance
- £350 St Edmund Hall Postgraduate College Grant for the [Causal Learning and Reasoning conference 2023](#)
- DST Group PhD Study Support Programme Award supporting five years of doctoral research at the University of Oxford on causal artificial intelligence for robust robot reasoning under uncertainty
- [Best Paper Award](#) at the ICRA 2021 [Robot Swarms in the Real World Workshop](#)
- Kwong Lee Dow Scholarship (University of Melbourne) awarded for excellence in leadership and academia
- Melbourne Global Scholars Award supporting a six-month international exchange at KTH Royal Institute of Technology, Stockholm, Sweden

ACADEMIC TALKS

- Invited talk: Causality for Confounded POMDP Robot Planning, IROS 2023 [Causality for Robotics workshop](#)
- Gave a series of Cognitive Robotics Group lab research overview presentations and demonstrations to international visitors at the Oxford Robotics Institute as part of the official [ICRA 2023 lab tours](#)

ACADEMIC SERVICE

- Assisted coordination of the ICRA 2023 [Multidisciplinary Approaches to Co-Creating Trustworthy Autonomous Systems workshop](#). Recorded workshop discussion minutes, coordinated breakout groups, social media coverage.
- Manuscript reviews: ICRA 2025-2026, IROS (2022-2025), KI - Künstliche Intelligenz Journal (2023), CoRL 2025 workshop on Resource-Rational Robot Learning

TEACHING & SUPERVISION

- Final Year Research Project Co-Supervisor, University of Oxford Masters of Engineering Science (2021–2024)
 - Co-supervised four research projects on causal discovery, causal Bayesian planning, and natural language counterfactual explanation methods for (1) robot manipulation tasks and (2) autonomous drones in mines
 - Wrote project proposals and on-boarding materials, performed candidate selection, and delivered educational tutorials. Provided guidance on research methods, project management, robotics, and causal ML expertise.
- Teaching Assistant, University of Oxford third-year Information Engineering Systems course (2021/22)
 - Taught three tutorial cohorts on image and digital signal processing, estimation, and inference
 - Consistently rated outstanding for quality of teaching and student support from student evaluations

SKILLS

Machine Learning & Generative Models	Deep learning, generative AI, multi-modal models, latent diffusion, transformers, representation learning, variational autoencoders (VAEs)
Causal & Probabilistic Modelling	Probabilistic generative causal modelling, causal Bayesian inference, counterfactual reasoning, human-aligned causal attribution, causally faithful relationship modelling, probabilistic programming, uncertainty-aware modelling, stochastic variational inference
Research Software & Tooling	Python, C++, PyTorch, Pyro probabilistic programming language, HuggingFace Transformers, PyGame (interactive game and environment development), PyBullet (physics-based simulation), experiment tracking (Weights & Biases), reproducible research workflows, Git, Linux-based research environments
Decision-Making & Reasoning Systems	Probabilistic planning under uncertainty, partially observable Markov decision processes (POMDPs), Monte Carlo Tree Search (MCTS), search algorithms, graphical models, knowledge representation and reasoning, interactive environment design for agent learning and evaluation, interpretable decision-making systems
Robotics & Autonomous Systems (Applied)	Autonomous systems design and deployment, simulation and verification, field trials, unmanned aerial systems (UAS), perception, estimation, control, simultaneous localisation and mapping (SLAM)
Professional & Research Skills	Research project leadership, supervision and mentorship, cross-disciplinary collaboration, stakeholder engagement, scientific communication and presentation

LEADERSHIP

- Organisational, community, and education leadership involving strategic planning, programme delivery, and execution of complex, multi-stakeholder initiatives, with responsibility for governance, compliance, reporting, and ethical conduct in scientific outreach contexts
- Executive Committee Member, Young Scientists of Australia (YSA), Melbourne Chapter (2008–2019)
 - Immediate Past President (2016–2019), President (2013–2016), Social Coordinator (2009–2012)
 - Led a 12-person executive committee responsible for organisational governance, strategic planning, budgeting, compliance, and reporting for a 200+ member scientific organisation
 - Planned, coordinated, and executed an annual programme of 12 science outreach and professional development events, managing schedules, resources, risk, and stakeholder coordination
 - Directed the delivery of five multi-day STEM outreach programmes annually, coordinating 40+ volunteer staff and ensuring adherence to safety, ethical, and organisational requirements
 - Initiated and delivered seven new organisational initiatives to expand member services, outreach reach, and long-term organisational sustainability
 - Managed formal relationships and collaborative activities with partner science organisations including CSIRO, the Royal Society of Victoria, ANZAAS, and National Science Week
- Federal and Victoria Committee Member, Australian and New Zealand Association for the Advancement of Science (ANZAAS) (2013–2018)
 - Contributed to national-level governance, programme delivery, and administration supporting ANZAAS’s mission to promote and communicate science across Australia
 - Co-organised multi-day residential *Youth ANZAAS* STEM outreach programmes for secondary school students (2014–2016)

PUBLICATIONS

- *Multiverse Mechanics: A Testbed for Learning Game Mechanics via Counterfactual Worlds*. **Ricardo Cannizzaro***, Robert Osazuwa Ness*, Yunshu Wu, Lars Kunze. Manuscript under review at the International Conference on Learning Representations (ICLR), 2026. [\[Paper\]](#)
- *COBRA-PPM: A Causal Bayesian Reasoning Architecture Using Probabilistic Programming for Robot Manipulation Under Uncertainty*. **Ricardo Cannizzaro**, Michael Groom, Jonathan Routley, Robert Osazuwa Ness, Lars Kunze. Proceedings of the 12th European Conference on Mobile Robots (ECMR), 2025. [\[Paper\]](#)
- *Aloft: Self-Adaptive Drone Controller Testbed*. Calum Imrie, Rhys Howard, Divya Thuremella, Nawshin Mannan Proma, Tejas Pandey, Paulina Lewinska, **Ricardo Cannizzaro**, Richard Hawkins, Colin Paterson, Lars Kunze, Victoria Hodge. Proceedings of the IEEE/ACM 19th Symposium on Software Engineering for Adaptive and Self-Managing Systems (SEAMS), 2024. [\[Paper\]](#)
- *CAR-DESPOT: Causally-Informed Online POMDP Planning for Robots in Confounded Environments*. **Ricardo Cannizzaro**, Lars Kunze. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023. [\[Paper\]](#)
- *Towards a Causal Probabilistic Framework for Prediction, Action-Selection & Explanations for Robot Block-Stacking Tasks*. **Ricardo Cannizzaro***, Jonathan Routley*, Lars Kunze. IEEE/RSJ IROS Workshop on Causality for Robotics, 2023. [\[Paper\]](#)
- *Towards Probabilistic Causal Discovery, Inference & Explanations for Autonomous Drones in Mine Surveying Tasks*. **Ricardo Cannizzaro***, Rhys Howard*, Paulina Lewinska, Lars Kunze. IEEE/RSJ IROS Workshop on Causality for Robotics, 2023. [\[Paper\]](#)

- *An Upper Confidence Bound for Simultaneous Exploration and Exploitation in Heterogeneous Multi-Robot Systems*. Ki Myung Brian Lee, Felix Kong, **Ricardo Cannizzaro**, Jennifer L Palmer, David Johnson, Chanyeol Yoo, Robert Fitch. IEEE International Conference on Robotics and Automation (ICRA), 2021. [\[Paper\]](#)
- *Decentralised Intelligence, Surveillance, and Reconnaissance in Unknown Environments with Heterogeneous Multi-Robot Systems*. Ki Myung Brian Lee, Felix H. Kong, **Ricardo Cannizzaro**, Jennifer L Palmer, David Johnson, Robert Fitch. IEEE ICRA Workshop on Robot Swarms in the Real World, 2021. [\[Paper\]](#)
- *ColMap: A Memory-Efficient Occupancy Grid Mapping Framework*. Alex Fisher, **Ricardo Cannizzaro**, Madeleine Cochrane, Chatura Nagahawatte, Jennifer L Palmer. *Robotics and Autonomous Systems*, 2021. [\[Paper\]](#)
- *Evaluation of LIDAR and X-Band Radar Sensors in a Particle-Dense Environment*. **Ricardo Cannizzaro**, Meaghan G Smith, Alexander W Blain, Thomas Baum, Kelvin Nicholson, Jennifer L Palmer. Proceedings of the Australian International Aerospace Congress (AIAC), 2019. [\[Paper\]](#)
- *Experimental Validation of a Random-Finite-Set-Based SLAM Algorithm for Micro UAS*. **Ricardo Cannizzaro**, Madeleine Cochrane, Lance Fang, Jennifer L Palmer. Australasian Conference on Robotics and Automation (ACRA), 2019. [\[Paper\]](#)
- *Data Ferrying with Swarming UAS in Tactical Defence Networks*. Robert Hunjet, Bradley Fraser, Thomas Stevens, Laura Hodges, Karina Mayen, Jan Carlo Barca, Madeleine Cochrane, **Ricardo Cannizzaro**, Jennifer L Palmer. IEEE International Conference on Robotics and Automation (ICRA), 2018. [\[Paper\]](#)
- *Source Localisation with a Bernoulli Particle-Filter-Based Bearings-Only Tracking Algorithm*. Jennifer L Palmer, **Ricardo Cannizzaro**, Branko Ristic, Thomas Cheah, Chatura Nagahawatte, John Gilbert, Sanjeev Arulampalam. Australasian Conference on Robotics and Automation (ACRA), 2015. [\[Paper\]](#)